



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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CSA 0508-DATABASE MANAGEMENT SYSTEMS

ANALYSIS OF ANSI/SPARC THREE-LEVEL DATABASE ARCHITECTURE

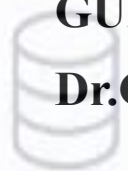
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DEPT:COMPUTER SCIENCE AND ENGINEERING



Data Storage



Data Security



Data Management



Data Analysis

The **ANSI/SPARC Three-Level Architecture** is a **database architecture** proposed by the **ANSI (American National Standards Institute)** and **SPARC (Standards Planning and Requirements Committee)** in 1975.

It is a framework used in **Database Management Systems (DBMS)** to organize how data is viewed, managed, and stored.

- **Why was it introduced?**

Before this architecture, if the database structure changed, every application using the database also had to be changed. This was difficult and time-consuming.

The ANSI/SPARC architecture was introduced to solve this problem by providing **data independence**, meaning changes at one level do not significantly affect the other levels.

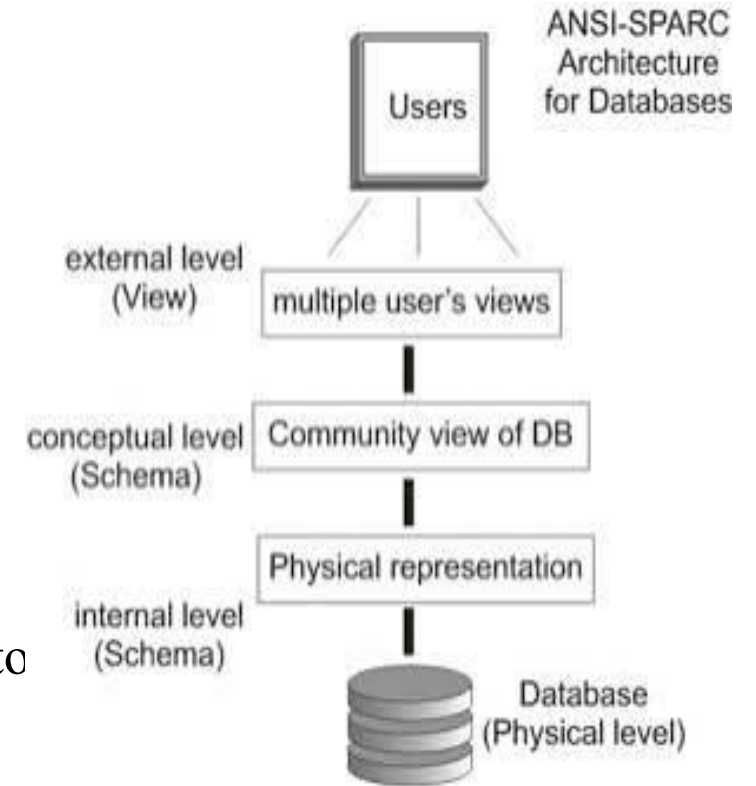
What is it?

Think of a database like a large library.

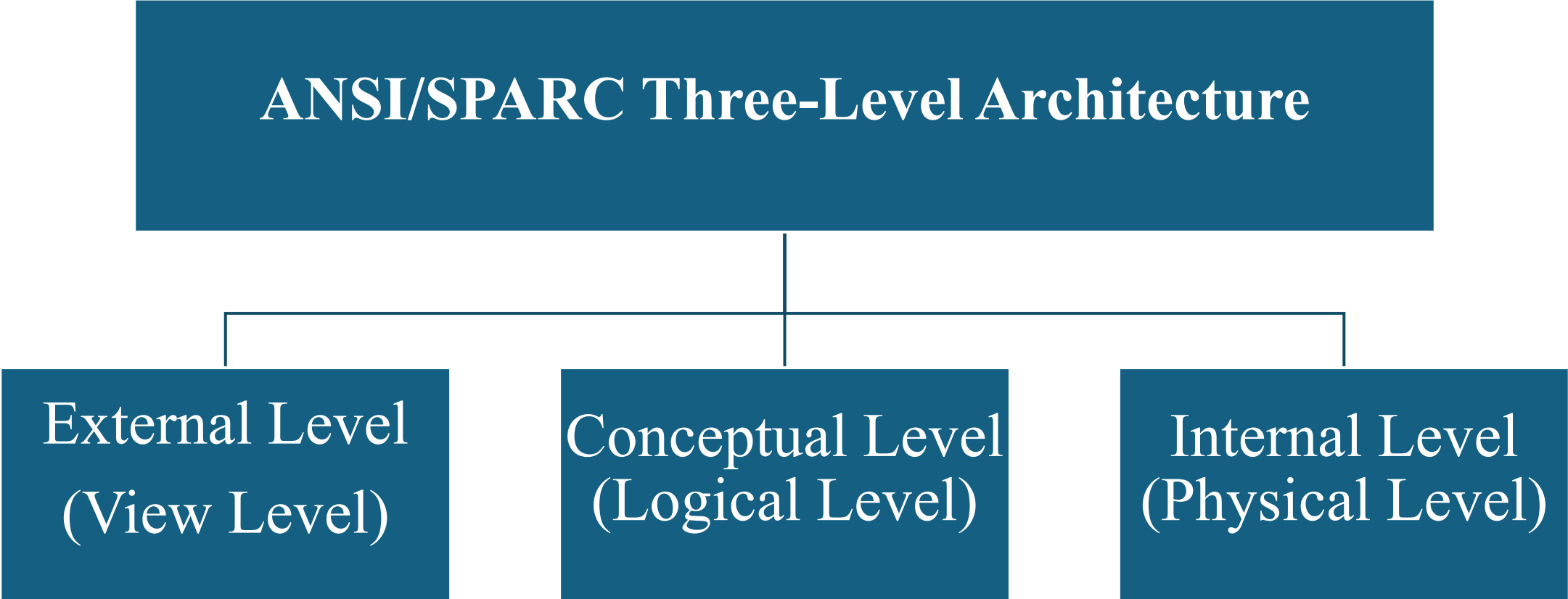
- **Students** can only see and borrow books.
- **Librarians** can add, remove, and update books.
- **The storage room** contains all the books arranged on shelves.

Each person interacts with the library differently, but they are all using the same library. Similarly, the ANSI/SPARC architecture allows different users to access the same database in different ways while hiding unnecessary details.

Its main purpose is to **separate the user's view of data from the way the data is physically stored.**



ANSI/SPARC Three-Level Architecture



```
graph TD; A[ANSI/SPARC Three-Level Architecture] --> B[External Level  
(View Level)]; A --> C[Conceptual Level  
(Logical Level)]; A --> D[Internal Level  
(Physical Level)];
```

The diagram illustrates the ANSI/SPARC Three-Level Architecture. It consists of a top-level box labeled "ANSI/SPARC Three-Level Architecture" which branches down to three sub-levels: "External Level (View Level)", "Conceptual Level (Logical Level)", and "Internal Level (Physical Level)".

External Level
(View Level)

Conceptual Level
(Logical Level)

Internal Level
(Physical Level)

1. External Level (View Level)

- The highest level of the architecture.
- Displays only the data required by a specific user or application.
- Different users can have different views of the same database.
- Ensures **security** and **data abstraction**.

Example:

In a college database:

- Students can view their marks and attendance.
- Faculty can view student records.
- Accounts staff can view fee details.

2. Conceptual Level (Logical Level)

- The middle level of the architecture.
- Describes the **overall logical structure** of the entire database.
- Defines tables, attributes, relationships, and constraints.
- Independent of physical storage details.

Example:

A college database contains tables such as:

- STUDENT
- COURSE
- FACULTY
- ENROLLMENT
- It defines how these tables are related.

3. Internal Level (Physical Level)

- The lowest level of the architecture.
- Describes how data is **physically stored** on storage devices.
- Includes file organization, indexes, storage blocks, and access methods.
- Managed by the DBMS and hidden from users.

Example:

The STUDENT table is stored as files on a hard disk or SSD, with indexes used to retrieve records efficiently.

Data Independence

Physical Data Independence

- Changes in physical storage do **not** affect the conceptual level.

Example:

- Change HDD to SSD
- Create an index
- Compress database files
- Users continue using the database normally.

Logical Data Independence

- Changes in the conceptual schema do **not** affect external views.

Example:

- Add Email column
- Add Department table
- Existing applications continue to work with little or no modification.

Real-Life Example

- Imagine **Instagram**.

External Level: You see your feed, profile, and messages.

Conceptual Level: Instagram organizes users, posts, comments, followers, and likes into related database tables.

Internal Level: The actual data is stored across servers, hard drives, and cloud storage using optimized storage techniques.

You never need to know where your photos are physically stored—you just access them through the app

CONCLUSION

The ANSI/SPARC Three-Level Architecture organizes a database into External, Conceptual, and Internal levels. It provides data abstraction, security, and data independence, making database systems more flexible, efficient, and easier to maintain.

THANKYOU!